

Trade Study — Home Connectivity Element, PD-RPM Remote Care System

This study selects the **home connectivity element** of the PD-RPM remote care system — function **F2.0 Buffer and Transport Data** in the Module 4 functional diagram, realized physically as WBS element 1.1.4. It is the right element to trade first because every operational requirement written in Module 3 depends on it: if a completed treatment never reaches the monitoring center, no amount of analytics recovers it. The study follows the twelve-step JHU construction process in order.

Step 1 — Define Objectives and Requirements

The decision. Select the home-to-monitoring-center connectivity approach that satisfies every operational requirement and is cost effective across a three-year service life. **Objectives.** Deliver every completed treatment session to the monitoring center without patient effort, fast enough to support a fifteen-minute alert path, in substantially all enrolled homes. **Governing requirements** carried forward from M3A1: OR-2 (automatic transfer within 30 minutes of session end), KPP-1 Alert Latency, KPP-3 Data Completeness, KPP-4 Patient Daily Interaction Time, and ON-2 (no added patient burden).

Step 2 — Identify Alternatives (at least 3)

Four alternatives are carried, exceeding the three-alternative minimum. They span an existing solution, current technology, and a combined approach, as the process directs.

	Alternative	Description
A1	Broadband Wi-Fi gateway	Small appliance bridges the cyclor and peripherals to the patient's existing home internet.
A2	Dedicated LTE-M cellular gateway	Carrier-managed cellular gateway supplied with the therapy, independent of household internet.
A3	Smartphone-tethered BLE application	Devices pair to the patient's own phone over Bluetooth; the phone relays to the cloud.
A4	Dual-path gateway (broadband primary, cellular failover)	Broadband when available, automatic cellular failover, single managed device.

Step 3 — Formulate Selection Criteria

Each criterion traces to at least one requirement. C4 had no corresponding requirement in the Module 3 specification, so one was created rather than scoring an untraceable criterion: **OR-6 — the connectivity element shall serve as the primary data path in at least 90 percent of enrolled patient homes.** This is added to the system specification. Following course convention, **cost is not a selection criterion**; it enters only through the cost-effectiveness selection function in Step 7.

	Selection criterion	Definition and units	Traces to
C1	Session Delivery Completeness	% of completed cyclor sessions ingested within 24 h	KPP-3 Data Completeness ($\geq 95\%$); OR-2
C2	Time to Data Availability	median minutes from session end to availability for detection	OR-2 (≤ 30 min); KPP-1 Alert Latency
C3	Patient Daily Interaction Time	seconds per day of patient action attributable to the link	KPP-4 (≤ 60 s); ON-2
C4	Deployable Coverage	% of enrolled homes where the approach serves as primary path	OR-6 ($\geq 90\%$) — created for this study

Step 4 — Weight the Criteria

Weights are derived by **Nth-root pairwise comparison** rather than assigned, using the 1–9 importance scale. Every pair of criteria is judged, the reciprocal fills the matrix below the diagonal, and the normalized fourth root of each row product gives the weight.

Pair	More important	Level	Basis for the judgment
C1 – C2	C1	3	Data that arrives late still supports trending; data that never arrives does not.
C1 – C3	C1	2	Both drive technique failure, but an unmonitored patient is the larger risk.
C1 – C4	C1	3	Coverage gaps can be bridged case by case; missing sessions cannot be recovered.
C2 – C3	C3	2	Burden is continuous and compounding; latency inside 30 min is clinically equivalent.
C2 – C4	Equal	1	Neither dominates: both constrain reach rather than clinical value directly.
C3 – C4	C3	2	A burdensome link loses the patient; an uncovered home is never enrolled on it.

	C1	C2	C3	C4	Row product	4th root	Weight
C1	1	3	2	3	18.0000	2.0598	0.455
C2	1/3	1	1/2	1	0.1667	0.6389	0.141
C3	1/2	2	1	2	2.0000	1.1892	0.263
C4	1/3	1	1/2	1	0.1667	0.6389	0.141

No criterion exceeds 50 percent (the largest is C1 at 45.5 percent) and none falls below 0.03, so neither the re-comparison check nor the drop check is triggered. **Weights are not ranks**, and none of these values was chosen directly.

Step 5 — Collect Data

Values are drawn from manufacturer specifications, carrier coverage data for the service region, and the home program's own installation records. Values failing a requirement are shown in red.

	Min req't	Max req't	A1	A2	A3	A4
C1 KPP	95	100	96.5	98.2	91	99.3
C2	—	30	12	18	26	11
C3 KPP	—	60	10	0	45	0
C4	90	100	78	94	83	97

KPP check. C1 is a KPP, and its threshold is binding: **A3 delivers 91 percent against a 95 percent KPP threshold and is therefore disqualified.** It is carried through the arithmetic for completeness but is excluded from selection. C4 is a requirement rather than a KPP, so A1 at 78 percent and A3 at 83 percent are not automatically disqualified — they may still have value — but the shortfall is carried into Step 9 as a risk.

Step 6 — Prepare Utility Functions

Utility functions convert raw values in four different units onto a common unitless 0–1 scale. Three are linear, since no stakeholder information justifies a different shape. C4 is piecewise linear with a knee at the OR-6 requirement, reflecting that coverage below the requirement is worth materially less than a proportional reading would suggest.

	Utility anchors	Shape	A1	A2	A3	A4
C1	0 at 95% (KPP threshold — below it the system is unusable), 1 at 100%	linear	0.30	0.64	0.00	0.86
C2	1 at 0 min, 0 at the 30 min ceiling in OR-2	linear, negative slope	0.60	0.40	0.13	0.63
C3	1 at 0 s, 0 at the 60 s ceiling in KPP-4	linear, negative slope	0.83	1.00	0.25	1.00
C4	0 at 70%, 0.5 at the 90% requirement, 1 at 100%	piecewise linear, knee at OR-6	0.20	0.70	0.33	0.85

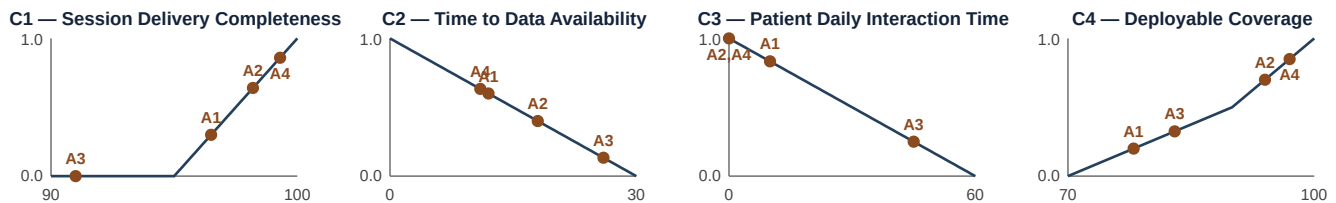


Figure 1. Utility functions with the four alternatives plotted. A3 falls to zero utility on C1 because it sits below the KPP threshold.

Step 7 — Evaluate Alternatives

Weighted utility is the product of weight and utility value; the operational utility function is their sum; the cost-effectiveness selection function is that sum divided by three-year unit cost, times 100.

Criteria	Wt.	A1 Raw	A1 Utility	A1 Wtd	A2 Raw	A2 Utility	A2 Wtd	A3 Raw	A3 Utility	A3 Wtd	A4 Raw	A4 Utility	A4 Wtd
C1	0.455	96.5	0.30	0.137	98.2	0.64	0.291	91	0.00	0.000	99.3	0.86	0.391
C2	0.141	12	0.60	0.085	18	0.40	0.056	26	0.13	0.019	11	0.63	0.089
C3	0.263	10	0.83	0.219	0	1.00	0.263	45	0.25	0.066	0	1.00	0.263
C4	0.141	78	0.20	0.028	94	0.70	0.099	83	0.33	0.046	97	0.85	0.120
Operational Utility Function				0.468			0.709			0.130			0.863
Cost (\$) 3-year unit cost				180			420			40			540
Cost-Effectiveness Selection Function				0.260			0.169			0.326			0.160

The two functions disagree. A4 has the highest operational utility at 0.863, while A3 has the highest cost-effectiveness at 0.326 — but A3 is disqualified on a KPP. Among the qualified alternatives A1 is the most cost-effective at 0.260 and A4 the least at 0.160, so the ranking reverses depending on which function governs the decision.

Step 8 — Perform Sensitivity Check

Each criterion weight is set to zero in turn and the operational utility function is recalculated, as the process directs. The question is whether the ranking is stable.

Weight set to zero	A1	A2	A3	A4	Utility ranking
none (baseline)	0.468	0.709	0.130	0.863	A4 > A2 > A1 > A3
C1	0.332	0.418	0.130	0.472	A4 > A2 > A1 > A3
C2	0.384	0.653	0.112	0.774	A4 > A2 > A1 > A3
C3	0.249	0.446	0.065	0.601	A4 > A2 > A1 > A3
C4	0.440	0.610	0.084	0.743	A4 > A2 > A1 > A3

The result is insensitive. The ranking A4 > A2 > A1 > A3 holds under every single-criterion zeroing, so small errors in any one weight do not change the outcome. Insensitivity is the desired finding: it means the conclusion is not an artifact of the weighting. The study is sensitive to something else — which of the two selection functions governs — and that is a decision about value, not a data error.

Step 9 — Make Necessary Adjustments

Decision makers make decisions; trade studies produce results. Three adjustments follow from the analysis. First, A3 is removed from consideration on the KPP failure, not on its score. Second, A1's apparent cost-effectiveness advantage (0.260 against A2's 0.169) is an artifact of an incomplete cost basis: at 78 percent deployable coverage it fails OR-6, and the remaining 22 percent of homes would require a second connectivity approach whose cost is not in the 180 dollar unit figure. Third, A2 and A4 are separated by cost rather than by capability, and A4's added utility is concentrated in C1, where A2 already clears the KPP.

Step 10 — Select Preferred Alternative

Select A2, the dedicated LTE-M cellular gateway. It is the only alternative that satisfies every requirement including the newly created OR-6, it clears both KPP thresholds with margin, it scores second on operational utility (0.709 against A4's 0.863), and it reaches zero patient daily interaction time, which protects the requirement most likely to drive technique failure. A4 is retained as the fallback for homes where cellular signal is marginal, recorded as a documented variant rather than a second procurement.

Step 11 — Document Decision

The decision record captures: the decision statement and objectives; the four alternatives and their sources; the four criteria with their requirement traces; the pairwise judgments and the resulting weights; the raw data with provenance; the utility function anchors and shapes; the completed trade matrix; the sensitivity results; and the assumptions — principally that carrier coverage data predicts in-home signal, and that the three-year cost basis is the right horizon. OR-6 is added to the system specification as a product of this study.

Step 12 — Execute Decision

Execution: issue OR-6 against the system specification; fix A2 in the physical architecture under WBS 1.1.4 and define the carrier agreement as a managed interface under WBS 1.5.2; qualify A4 as the marginal-signal variant; and set a re-trade trigger — if measured deployable coverage falls below 90 percent in any region, or if observed session delivery completeness falls below the KPP for two consecutive months, this study is reopened rather than waived.

Systems Analysis — Fifth Discipline Archetype

Archetype 1, Limits to Growth. *Analyze problem:* the home program intends to grow census, and connectivity is the constraint it will meet first. *Conceptualize growth limits:* each added patient consumes deployable coverage, and the broadband-only alternative exhausts it at roughly 78 percent of homes, at which point growth stalls for reasons no one attributes to the connectivity choice made years earlier. *Conduct mental exercises:* with A1 the limit binds immediately; with A2 it moves to nurse triage capacity, which is a limit the program can actually buy its way out of. *Catalogue key variable factors:* deployable coverage, installation labor per patient, nurse capacity, alert volume. *Investigate solution options:* the leverage is in removing the limit rather than pushing harder on enrollment — which is exactly what selecting on coverage rather than on unit cost does.